Diagnostic procedures

Clinical presentation and physical examination

The diagnosis of systemic sclerosis (SSc) is challenging due to the heterogeneity of disease manifestations and variable disease course. In 1980, the American College of Rheumatology (ACR) published preliminary classification criteria for patients with established disease. A subclassification developed by LeRoy et al. has become the most widely used system in clinical practice and serves as the foundation for many global registries (Table 3).

It is now widely accepted that **CREST syndrome** (calcinosis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, telangiectasia) and **systemic sclerosis sine scleroderma** fall within the spectrum of **limited cutaneous SSc**.

For patients with very early disease—also referred to as very early/early SSc, pre-SSc, or undifferentiated connective tissue disease—there are no universally accepted diagnostic criteria. Among individuals presenting with **Raynaud's phenomenon**, **nail fold capillaroscopic abnormalities**, and/or **SSc-specific autoantibodies** (e.g., anti-centromere, anti-topoisomerase-1), approximately two-thirds will develop definite SSc within 5 years, and nearly **80%** will progress to definite disease over the long term. In contrast, those without a scleroderma pattern on capillaroscopy or SSc-specific antibodies have a very low risk of progression to SSc (**1.8%** during long-term follow-up).

Skin manifestations

Raynaud's phenomenon is present in more than **90%** of patients with SSc. It most commonly affects the hands, less frequently the feet, and may also involve the tongue, ears, and nose. Typical triggers include cold exposure and emotional stress.

At disease onset—particularly in the diffuse cutaneous form—patients often experience prolonged swelling of the fingers and hands ("**puffy hands**"). This is followed by progressive skin thickening (sclerosis), leading to **sclerodactyly**, **dermatogenic contractures**, **perioral plication**, **microstomia**, and a **mask-like facial appearance**.

Additional cutaneous features may include: - Hair loss - Reduced sweating (hypohidrosis) - Hyperpigmentation or depigmentation - Severe pruritus

In later stages, skin fibrosis of the trunk and proximal extremities may stabilize or even improve, while internal organ involvement can progress.

Digital ulcers are a common complication: - **15–25%** of SSc patients have active digital ulcers - **Approximately 35%** have current or a history of digital ulcers (rates vary across centers)

Fingertip ulcers are typically ischemic in origin, whereas ulcers over the extensor surfaces of the interphalangeal joints result from a combination of poor perfusion, taut fibrotic skin, and trauma. Complications of digital ulcers include: - Secondary infection - Osteomyelitis - Acro-osteolysis - Gangrene

Calcinosis cutis—deposits of calcium carbonate in the subcutaneous tissue—occurs in all SSc subtypes and predominantly affects acral areas (fingers, elbows, knees). These deposits may cause pain, skin erosion, and ulceration.

Musculoskeletal manifestations

Arthralgia and **musculoskeletal pain** are among the most common complaints in SSc and may contribute to secondary **fibromyalgia**.

Tendon friction rubs, heard on auscultation over tendons (especially in the hands), are a characteristic sign of active inflammation and are associated with more progressive disease.

Muscle weakness and mild elevations in **serum creatine kinase (CK)** levels are relatively common and may suggest an **SSc-myositis overlap syndrome**.

Inflammatory arthritis occurs in up to **10%** of patients and should raise suspicion for an **SSc–rheumatoid arthritis overlap syndrome**.

Pulmonary manifestations

Interstitial lung disease (ILD) affects up to **65%** of SSc patients, with varying degrees of severity. The typical radiological pattern is **bibasilar** and most commonly corresponds to **non-specific interstitial pneumonitis (NSIP)**.

High-resolution computed tomography (HRCT) of the chest is more sensitive than pulmonary function tests in detecting early ILD and is considered essential in the evaluation of SSc-related lung involvement.